

Currency Analytics Platform: Executive Report

A Cloud-Native Solution for Automated FX Intelligence

1.0 Executive Summary

The Currency Analytics Platform is a fully automated, cloud-native system that transforms raw foreign exchange data from the Frankfurter API (<https://api.frankfurter.dev/>) into actionable trading insights and risk intelligence.

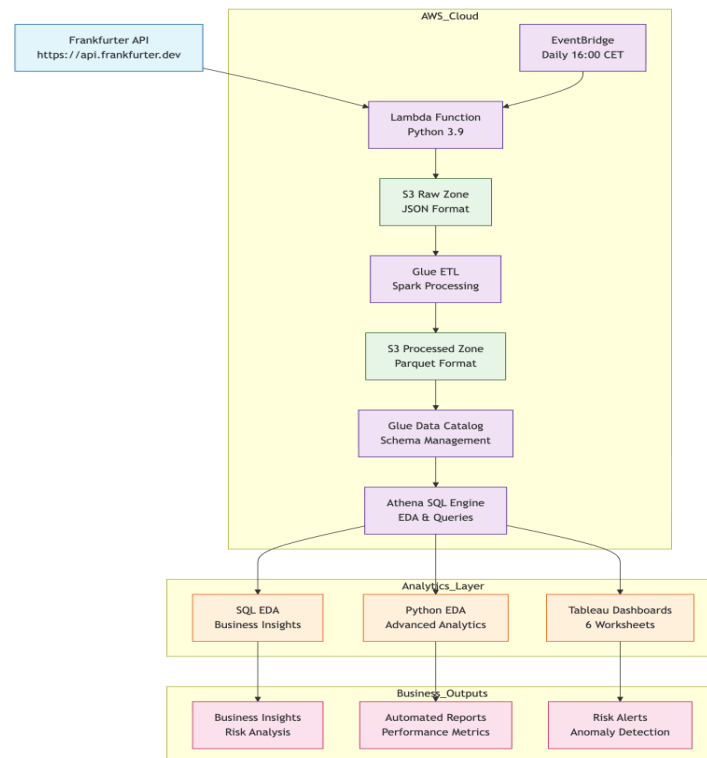
Manual currency analysis is time-consuming, inefficient, and prone to error. This platform solves that by providing traders, analysts, and portfolio managers with real-time, data-driven analytics.

2.0 Architecture Overview

The platform is built on a fully automated, serverless AWS pipeline, ensuring scalability, cost-efficiency, and minimal operational overhead.

2.1 System Architecture Diagram

(A visual representation of the complete system flow, from data ingestion to visualization.)



2.2 Technical Implementation & Security

Layer	Technology	Purpose
Ingestion	AWS Lambda (Python)	Daily API data collection
Storage	Amazon S3	Data Lake (Raw & Processed)
Processing	AWS Glue (Spark)	ETL & Metric Calculation
Catalog	AWS Glue Catalog	Schema Management
Crawlers	AWS Glue Crawlers	Auto-discover Raw Data Schema
Query	AWS Athena	SQL Analytics Engine
Scheduling	EventBridge	Automated Daily Pipeline
Visualization	Tableau	Interactive Dashboards

2.3 Data Pipeline Steps

Phase 1: Data Ingestion (Daily)

1) Trigger: EventBridge daily at 16:00 CET.

aws

Search

(Option+S)

Asia Pacific (Mumbai)

Anusha Shankar

Amazon EventBridge

Schedules

DailyCurrencyDataIngestion

Disable

Edit

Delete

Dashboard

Developer resources

Buses

Pipes

Scheduler

Integration

Schema registry

Documentation

Schedule detail

Schedule name

DailyCurrencyDataIngestion

Status

Enabled

Description

Trigger daily currency data fetch at 16:00 CET

Schedule group name

default

Schedule ARN

arn:aws:scheduler:ap-south-1:2588993894758:schedule/default/DailyCurrencyDataIngestion

Action after completion

NONE

Schedule start time

-

Schedule end time

-

Execution time zone

Asia/Calcutta

Flexible time window

15 minute

Created date

Oct 17, 2025, 15:27:08 (UTC+05:30)

Last modified date

Oct 17, 2025, 15:27:08 (UTC+05:30)

Schedule

Target

Retry policy

Dead-letter queue

Encryption

Cron expression

Info

015* * ? *

MinutesHoursDay of monthMonthDay of weekYear

Copy cron expression

Next 10 trigger date

Date and time are displayed in the selected time zone for which this schedule is set in UTC format, e.g. 'Wed, Nov 9, 2022 09:00 (UTC - 08:00)'

Mon, 20 Oct 2025 15:00:00 (UTC+05:30)

Tue, 21 Oct 2025 15:00:00 (UTC+05:30)

Wed, 22 Oct 2025 15:00:00 (UTC+05:30)

Thu, 23 Oct 2025 15:00:00 (UTC+05:30)

Fri, 24 Oct 2025 15:00:00 (UTC+05:30)

Sat, 25 Oct 2025 15:00:00 (UTC+05:30)

Sun, 26 Oct 2025 15:00:00 (UTC+05:30)

Mon, 27 Oct 2025 15:00:00 (UTC+05:30)

Tue, 28 Oct 2025 15:00:00 (UTC+05:30)

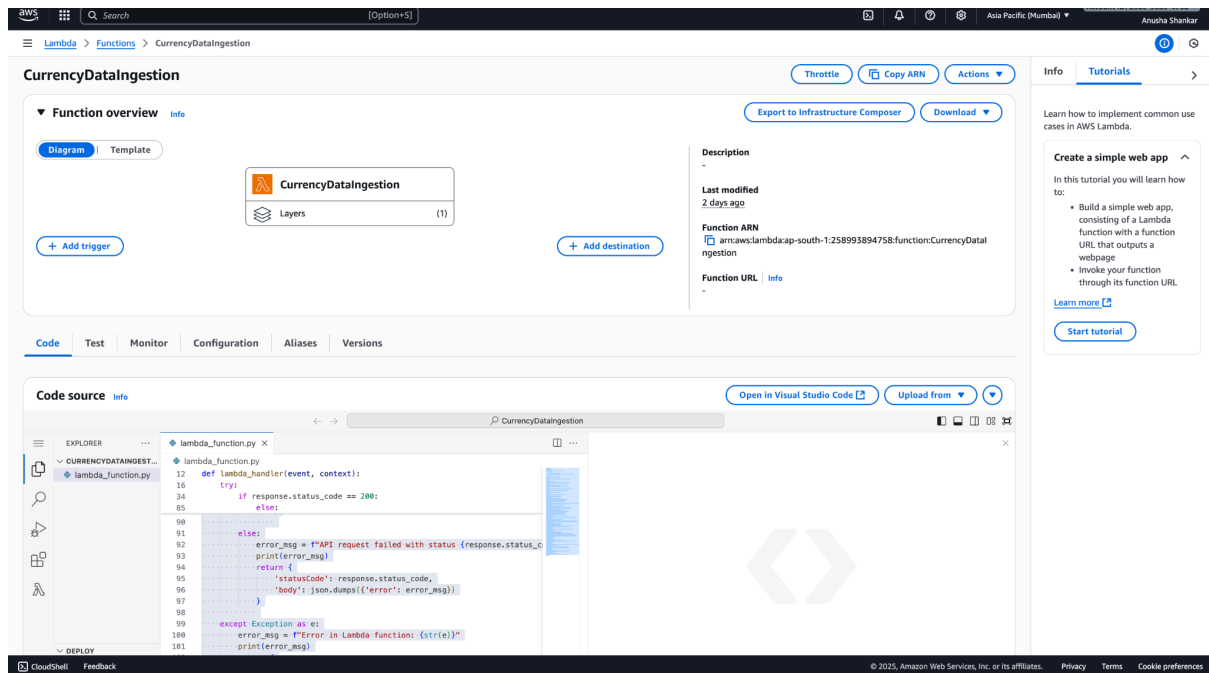
Wed, 29 Oct 2025 15:00:00 (UTC+05:30)

CloudShell

Feedback

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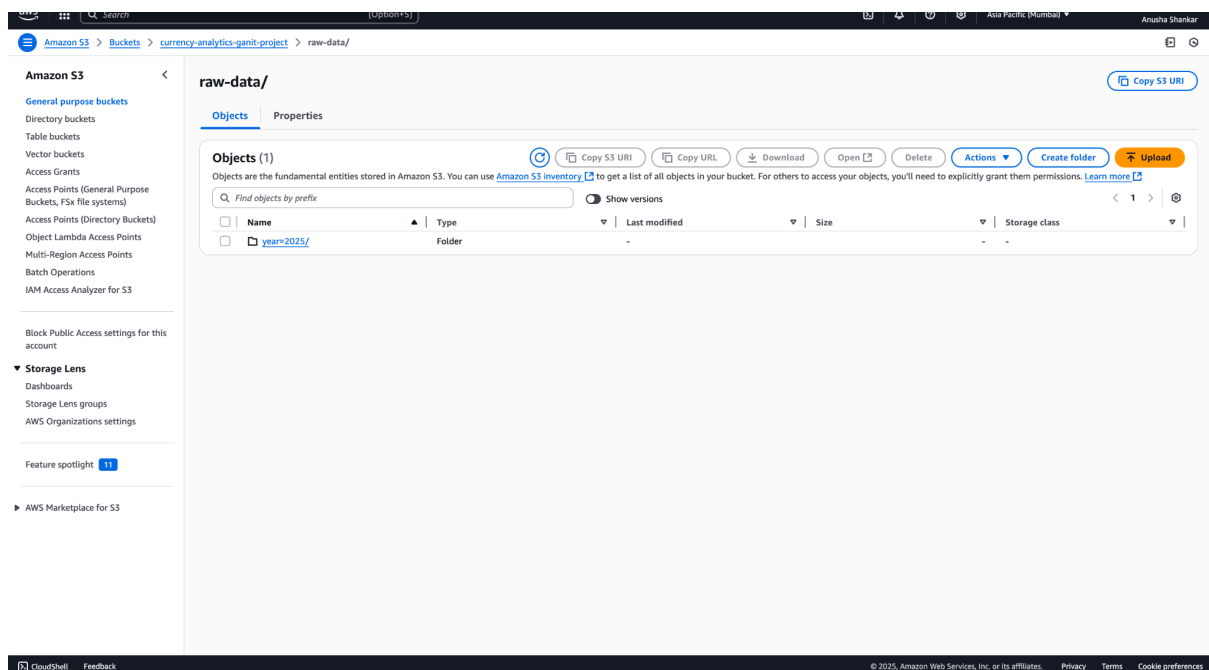
2) Process: Lambda function fetches historical rates from Frankfurter API.



The screenshot shows the AWS Lambda console for the 'CurrencyDataIngestion' function. The 'Function overview' tab is active, displaying the function name, layers, and a description. The 'Code source' tab is also visible, showing the Python code for the lambda_handler function. The code fetches historical rates from the Frankfurter API and stores them in S3.

```
def lambda_handler(event, context):
    try:
        if response.status_code == 200:
            # ... (code for fetching rates) ...
        else:
            error_msg = f"API request failed with status {response.status_code}"
            print(error_msg)
            return {
                'statusCode': response.status_code,
                'body': json.dumps({'error': error_msg})
            }
    except Exception as e:
        error_msg = f"Error in Lambda function: {str(e)}"
        print(error_msg)
```

3) Output: JSON files stored in S3, partitioned by year/month.

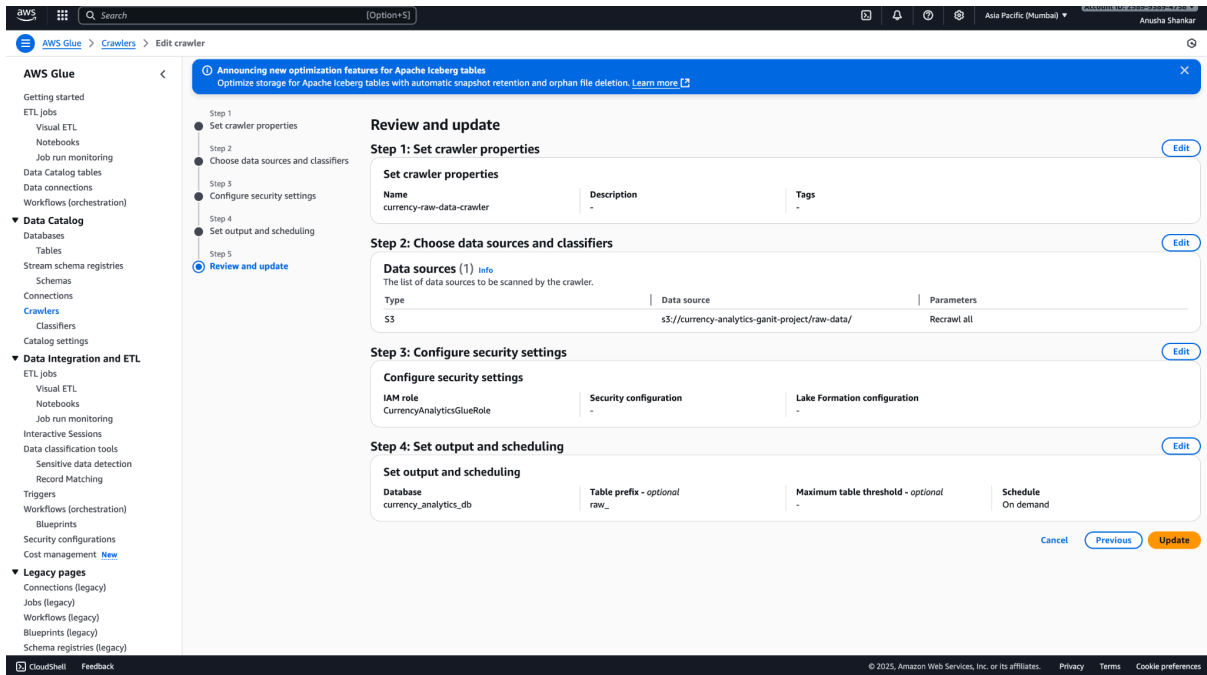


The screenshot shows the Amazon S3 console for the 'raw-data/' bucket. The 'Objects' tab is active, displaying a list of objects. The objects are JSON files stored in S3, partitioned by year/month.

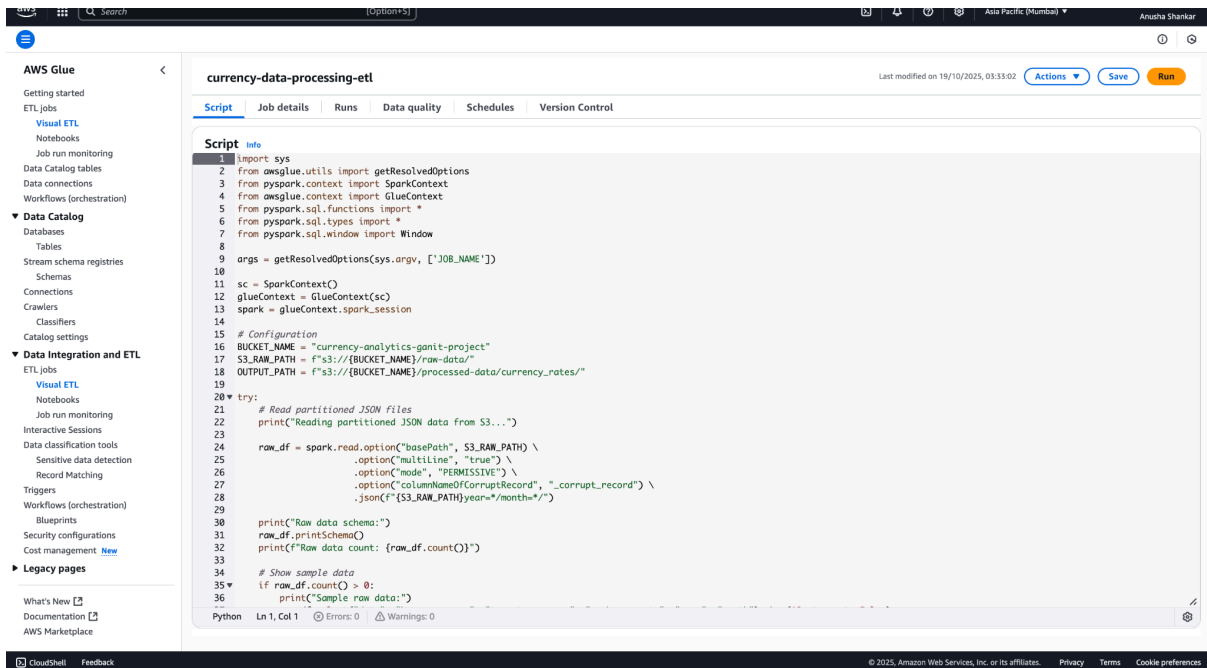
Name	Type	Last modified	Size	Storage class
year=2025/	Folder	-	-	-

Phase 2: Data Processing (Automated)

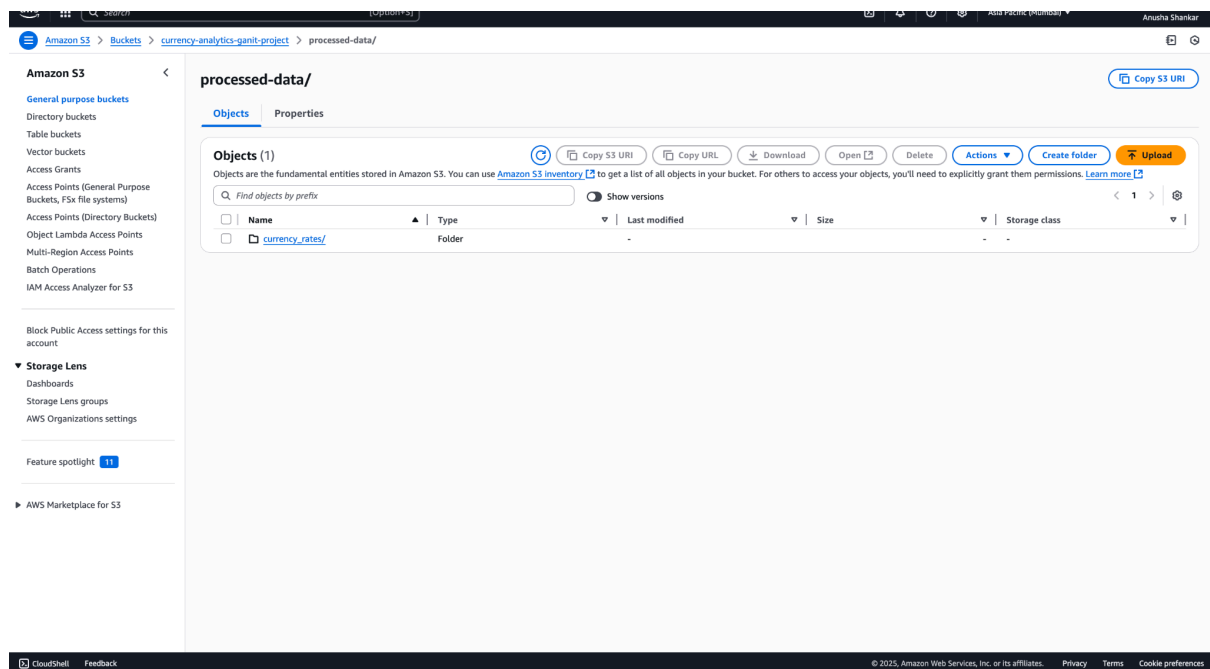
1) Process: Glue Crawler catalogs raw data



2) Glue ETL job cleans data, calculates metrics (returns, volatility), and converts it to Parquet format.

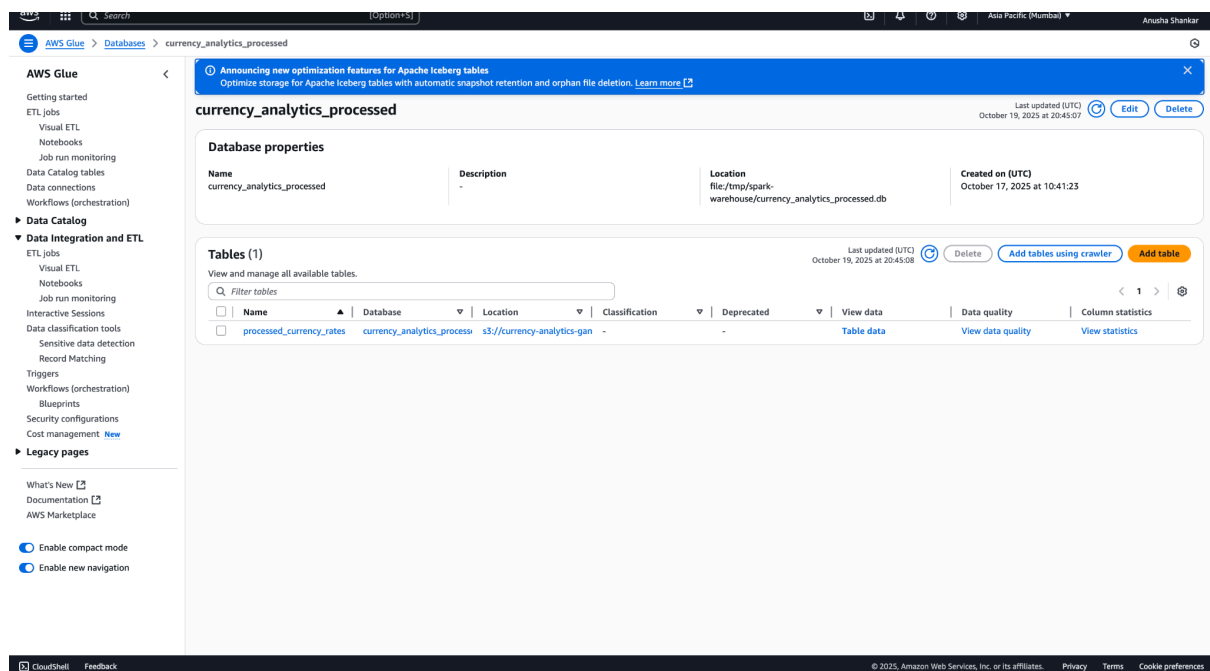


3) Output: Optimized, processed data stored in S3.



Phase 3: Data Catalog & Querying

1) Process: Glue Catalog makes processed data available as tables.



2) Access: Athena provides a SQL interface for ad-hoc querying and analysis.

Phase 4: Analytics Execution (Parallel)

1) Path A: SQL EDA Engine for core business insights.

- 2) Path B: Python EDA Engine for advanced statistical analysis.
- 3) Path C: Tableau for interactive visualization.

Phase 5: Insight Delivery

- 1) Outputs: Automated risk alerts, performance reports, correlation insights, and temporal pattern signals.

3.0 Key Analytical Findings

3.1 Critical Risks Identified:

- 1) Extreme Volatility: TRY (Turkish Lira), BRL (Brazilian Real), and MXN (Mexican Peso) form a high-risk cluster.
- 2) Unusual Behavior: JPY (Japanese Yen) is exhibiting atypical volatility for a major currency, requiring immediate investigation.
- 3) Regional Risk Clusters: Latin American currencies show high internal correlation and volatility.

3.2 Trading Opportunities:

- 1) Monday Effect: The strongest trading day, with an average return of +0.036%.
- 2) Friday Weakness: The worst day, with an average return of -0.007%; avoid new positions.
- 3) Best Risk-Adjusted Returns: CAD (Canadian Dollar) and SGD (Singapore Dollar) offer stability and favorable returns.

3.3 Market Patterns:

- 1) Asian Block Correlation: Currencies like SGD, HKD, CNY, and INR are highly correlated (>0.8), limiting hedging opportunities within the region.
- 2) Regional Linkages: High inter-region correlations reduce the effectiveness of geographic diversification.

4.0 Analytics & Intelligence Layer

4.1 SQL EDA Engine: Core Business Insights

This engine powers the foundational analytics, delivering clear, query-based insights.

- 1) Basic Data Overview

Scope: Comprehensive analysis of 21,481 records across 31 unique currencies

The screenshot shows the Amazon Athena query editor interface. The SQL query is as follows:

```
1 SELECT
2   COUNT(*) as total_records,
3   COUNT(DISTINCT target_currency) as unique_currencies,
4   MIN(trade_date) as earliest_date,
5   MAX(trade_date) as latest_date,
6   AVG(exchange_rate) as avg_exchange_rate
7 FROM currency_analytics_processed.processed_currency_rates;
```

The query results are displayed in a table with the following columns: #, total_records, unique_currencies, earliest_date, latest_date, and avg_exchange_rate. The results show 21,481 total records, 31 unique currencies, an earliest date of 2022-12-30, a latest date of 2025-10-17, and an average exchange rate of 661.1066560579126.

#	total_records	unique_currencies	earliest_date	latest_date	avg_exchange_rate
1	21481	31	2022-12-30	2025-10-17	661.1066560579126

Timeframe: Data spans from December 30, 2022, to October 17, 2025, providing a robust multi-year view

Baseline: The average exchange rate across the entire dataset is 661.11

2) Volatility Ranking

The screenshot shows the Amazon Athena query editor interface. The SQL query is as follows:

```
1 SELECT
2   target_currency,
3   AVG(volatility_30d) as avg_volatility,
4   MAX(daily_return) as max_daily_gain,
5   MIN(daily_return) as max_daily_loss
6 FROM currency_analytics_processed.processed_currency_rates
7 WHERE volatility_30d IS NOT NULL
8 GROUP BY target_currency
9 ORDER BY avg_volatility DESC
10 LIMIT 10;
```

The query results are displayed in a table with the following columns: #, target_currency, avg_volatility, max_daily_gain, and max_daily_loss. The results show the top 10 most volatile currencies, with BRL having the highest average volatility and maximum daily gain.

#	target_currency	avg_volatility	max_daily_gain	max_daily_loss
1	BRL	0.7499217227705629	4.148821852953543	-3.30909257671994
2	MXN	0.7167023824430493	5.155490767735665	-3.1074470012583455
3	ZAR	0.6698314065749075	4.2180660565061805	-1.8547242165040678
4	ILS	0.6638365580002574	3.1482684523511923	-2.183115338822857
5	JPY	0.5675277656283336	2.293929712460066	-3.3401499659168463
6	TRY	0.565373946096179	7.903899721448479	-2.9141730141458204
7	NOK	0.5218793481675671	2.3370286350211846	-2.596194503171236
8	AUD	0.4746117527933335	3.7253535415864378	-2.6425969224076984
9	NZD	0.46537172647924185	2.971596576888748	-2.2638092363416806
10	USD	0.4583794042848805	2.7214662595723844	-1.8537212076718361

Identified the top 10 most volatile currencies, highlighting significant risk concentrations and extreme price movers:

- BRL (Brazilian Real): 0.75% avg volatility | Extreme single-day gain: +4.15%
- MXN (Mexican Peso): 0.72% avg volatility | Highest single-day gain in dataset: +5.16%
- ZAR (South African Rand): 0.67% avg volatility
- ILS (Israeli Shekel): 0.66% avg volatility
- JPY (Japanese Yen): 0.57% avg volatility, Exhibiting unusual volatility for a major currency
- TRY (Turkish Lira): 0.57% avg volatility | Most extreme swing, with a +7.90% daily gain
- NOK (Norwegian Krone): 0.52% avg volatility
- AUD (Australian Dollar): 0.47% avg volatility
- NZD (New Zealand Dollar): 0.47% avg volatility
- USD (US Dollar): 0.46% avg volatility

3) Performance Analysis

The screenshot displays the Amazon Athena Query Editor interface. The SQL query is as follows:

```

1 SELECT
2   day_of_week,
3   AVG(daily_return) as avg_return,
4   AVG(volatility_30d) as avg_volatility,
5   COUNT(*) as trading_days
6 FROM currency_analytics_processed.processed_currency_rates
7 GROUP BY day_of_week
8 ORDER BY
9   CASE day_of_week
10    WHEN 'Mon' THEN 1
11    WHEN 'Tue' THEN 2
12    WHEN 'Wed' THEN 3
13    WHEN 'Thu' THEN 4
14    WHEN 'Fri' THEN 5
15    WHEN 'Sat' THEN 6
  
```

The query results are displayed in a table with the following columns: #, day_of_week, avg_return, avg_volatility, and trading_days. The results are for Monday through Friday.

#	day_of_week	avg_return	avg_volatility	trading_days
1	Mon	0.03550993875851198	0.4052322500781184	4200
2	Tue	0.013505608207426472	0.40766710491034314	4350
3	Wed	0.011325041959521049	0.40802443842288255	4290
4	Thu	0.01833603343545195	0.4061362056028969	4320
5	Fri	-0.006668708064810793	0.40309584504132445	4321

Uncovered a powerful and consistent day-of-week effect with direct tactical implications for traders:

- Monday: +0.036% avg return ← STRONGEST DAY (Optimal for opening new long positions)
- Thursday: +0.018% avg return
- Tuesday: +0.014% avg return
- Wednesday: +0.011% avg return
- Friday: -0.007% avg return ← WEAKEST DAY (Avoid new positions; implement hedging for weekend risk)

4) Correlation Analysis

The screenshot displays the Amazon Athena Query Editor interface. The top navigation bar includes 'Amazon Athena', 'Query editor tabs', and a 'Workgroup' dropdown set to 'primary'. Below the navigation bar, there are tabs for 'Editor', 'Recent queries', 'Saved queries', and 'Settings'. The main editor area contains a SQL query for correlation analysis, which is partially visible. The query starts with a CTE named 'correlations' and includes a CASE statement to categorize correlation strengths. Below the query, there are buttons for 'Run', 'Explain', 'Cancel', 'Clear', and 'Create'. The 'Query results' section shows a green bar indicating the query is 'Completed'. Below this, the 'Results (116)' section is visible, showing 'Time in queue: 55 ms', 'Run time: 997 ms', and 'Data scanned: 341.57 KB'. There are also buttons for 'Copy' and 'Download results CSV'.

```

1 WITH correlations AS (
2   SELECT a.target_currency as currency1,
3          b.target_currency as currency2,
4          CORR(a.daily_return, b.daily_return) as correlation,
5          COUNT(*) as overlapping_days
6   FROM currency_analytics_processed.processed_currency_rates a
7   JOIN currency_analytics_processed.processed_currency_rates b ON a.trade_date = b.trade_date
8   AND a.target_currency < b.target_currency
9   GROUP BY a.target_currency,
10            b.target_currency
11   HAVING COUNT(*) > 50
12 )
13 SELECT currency1,
14        currency2,
15        ROUND(correlation, 4) as correlation,
16        overlapping_days,
17        CASE
18          WHEN correlation > 0.8 THEN 'VERY_STRONG_POSITIVE'
19          WHEN correlation > 0.6 THEN 'STRONG_POSITIVE'
20          WHEN correlation > 0.4 THEN 'MODERATE_POSITIVE'
21          WHEN correlation > 0.2 THEN 'WEAK_POSITIVE'
22          WHEN correlation > -0.2 THEN 'NEUTRAL'
23          WHEN correlation > -0.4 THEN 'WEAK_NEGATIVE'
24          WHEN correlation > -0.6 THEN 'MODERATE_NEGATIVE'
25          WHEN correlation > -0.8 THEN 'STRONG_NEGATIVE' ELSE 'VERY_STRONG_NEGATIVE'
26        END as correlation_strength
27 FROM correlations
28 WHERE ABS(correlation) > 0.3 -- Only show meaningful correlations
29 ORDER BY ABS(correlation) DESC
SQL Ln 27, Col 18

```

Run Explain Cancel Clear Create

Query results Query status

Completed Time in queue: 55 ms Run time: 997 ms Data scanned: 341.57 KB

Results (116) Copy Download results CSV

The screenshot displays the Amazon Athena console interface. At the top, there's a navigation bar with 'Amazon Athena' and 'Query editor tabs'. Below this, a toolbar contains buttons for 'Run', 'Explain', 'Cancel', 'Clear', and 'Create'. The main area shows 'Query results' for a completed query. A status bar indicates 'Time in queue: 55 ms', 'Run time: 997 ms', and 'Data scanned: 341.57 KB'. The results are presented in a table with 7 columns: #, currency1, currency2, correlation, overlapping_days, and correlation_strength. The table contains 13 rows of data, showing various currency pairs and their correlation strengths.

#	currency1	currency2	correlation	overlapping_days	correlation_strength
1	HKD	USD	0.9926	715	VERY_STRONG_POSITIVE
2	INR	USD	0.908	715	VERY_STRONG_POSITIVE
3	HKD	INR	0.8977	715	VERY_STRONG_POSITIVE
4	CNY	USD	0.8682	715	VERY_STRONG_POSITIVE
5	CNY	HKD	0.8654	715	VERY_STRONG_POSITIVE
6	CNY	SGD	0.8369	715	VERY_STRONG_POSITIVE
7	AUD	NZD	0.8227	715	VERY_STRONG_POSITIVE
8	CNY	INR	0.814	715	VERY_STRONG_POSITIVE
9	SGD	USD	0.8058	715	VERY_STRONG_POSITIVE
10	HKD	SGD	0.8034	715	VERY_STRONG_POSITIVE
11	INR	SGD	0.7818	715	STRONG_POSITIVE
12	MYR	SGD	0.7523	715	STRONG_POSITIVE
13	IDR	SGD	0.7003	715	STRONG_POSITIVE

Mapped key currency relationships, revealing distinct regional blocks critical for hedging and diversification strategies:

- Very Strong Positive Correlations (HKD-USD: 0.9926, INR-USD: 0.9080)
- Trading Implication: These pairs move almost in lockstep, offering very limited arbitrage but simple, effective hedging
- Latin American Link: BRL-MXN: 0.5043 (Moderate correlation within the high-risk cluster)
- Asian Block: SGD, HKD, CNY, INR, MYR all show strong inter-correlations (>0.6), forming a cohesive regional bloc
- Scandinavian Pair: NOK-SEK: 0.5296

5) Regional & Cluster Analysis

The screenshot shows the Amazon Athena Query Editor interface. The SQL query is as follows:

```

1 WITH regional_groups AS (
2   SELECT
3     trade_date,
4     target_currency,
5     daily_return,
6     CASE
7       WHEN target_currency IN ('USD', 'CAD', 'MXN') THEN 'NORTH_AMERICA'
8       WHEN target_currency IN ('EUR', 'GBP', 'CHF', 'SEK', 'NOK') THEN 'EUROPE'
9       WHEN target_currency IN ('JPY', 'CNY', 'KRW', 'SGD') THEN 'ASIA_PACIFIC'
10      WHEN target_currency IN ('AUD', 'NZD') THEN 'OCEANIA'
11      ELSE 'OTHER'
12    END as currency_region
13  FROM currency_analytics_processed.processed_currency_rates
14  WHERE target_currency != 'EUR' -- Base currency
15 )
16
17 SELECT
18   a.currency_region as region1,
19   b.currency_region as region2,
20   CORR(a.daily_return, b.daily_return) as inter_region_correlation,
21   COUNT(*) as overlapping_days
22 FROM regional_groups a
23 JOIN regional_groups b
24   ON a.trade_date = b.trade_date
25  AND a.target_currency != b.target_currency
26  AND a.currency_region < b.currency_region
27 GROUP BY a.currency_region, b.currency_region
28 ORDER BY ABS(inter_region_correlation) DESC;

```

The query execution status is "Completed". The results show 15 rows. The time in queue is 54 ms, run time is 1.291 sec, and data scanned is 341.94 KB.

The screenshot shows the results of the SQL query. The results are displayed in a table with 15 rows. The columns are: #, region1, region2, inter_region_correlation, and overlapping_days.

#	region1	region2	inter_region_correlation	overlapping_days
1	LATIN_AMERICA	NORTH_AMERICA	0.38827756004103975	2145
2	EUROPE	OCEANIA	0.3470898564708716	5720
3	LATIN_AMERICA	OCEANIA	0.3292946362451989	1430
4	NORTH_AMERICA	OCEANIA	0.2611698519198233	4290
5	NORTH_AMERICA	OTHER	0.24675589516589908	34323
6	ASIA_PACIFIC	NORTH_AMERICA	0.23274886870259892	8580
7	ASIA_PACIFIC	OTHER	0.2211953738544378	45764
8	ASIA_PACIFIC	OCEANIA	0.19669644369940728	5720
9	LATIN_AMERICA	OTHER	0.189565804270177	11441
10	OCEANIA	OTHER	0.17095089910552932	22882
11	ASIA_PACIFIC	LATIN_AMERICA	0.15773480786749014	2860
12	EUROPE	LATIN_AMERICA	0.12568170353081112	2860
13	ASIA_PACIFIC	EUROPE	0.10470582616694056	11440
14	EUROPE	NORTH_AMERICA	0.09588136252485464	8580
15	EUROPE	OTHER	0.06485054436440969	45764

Quantified the strength of inter-market linkages, directly impacting global diversification effectiveness:

- Europe & Oceania: 0.347
- Asian Developed Currencies: High internal correlation (0.415), suggesting limited diversification benefits within the group
- Emerging Markets: Moderate internal correlation (0.298), indicating they often move together as an asset class

- Major Developed Currencies: Lower internal correlation (0.241), offering better diversification among themselves

6) Monthly Performance Trends

SQL: Ln 10, Col 39

Run again Explain Cancel Clear Create

Query results Query status

Completed Time in queue: 57 ms Run time: 748 ms Data scanned: 441.95 KB

Results (31)

Search rows

#	year	month	target_currency	avg_rate	avg_daily_return	avg_volatility
29	2025	10	TRY	34.54978839160841	0.1269608874718109	0.5665301485909134
17	2025	10	JPY	160.25832167832186	0.03328013504938634	0.5685386822044523
18	2025	10	KRW	1483.3874405594406	0.029759552250542956	0.42315656766096355
22	2025	10	NZD	1.815911328671327	0.02770473609067489	0.46620648913426743
13	2025	10	IDR	17274.52867132867	0.022954754136052608	0.4082987936135082
15	2025	10	INR	91.98781818181811	0.021924065260943287	0.42966819701126524
1	2025	10	AUD	1.666037762377625	0.01984710760671579	0.475460957960084
3	2025	10	BRL	5.813993846153853	0.019491750897740714	0.7512470750742456
23	2025	10	PHP	61.93853146853148	0.019286681824084943	0.42593569478246135
4	2025	10	CAD	1.4982608391608387	0.018195449692160717	0.3538623922695426

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- Top Performers: TRY (+0.127% avg daily return), JPY (+0.033%), KRW (+0.030%)
- Stable Low-Risk Options: DKK (0.014% volatility), SGD (0.257% volatility), CAD (0.354% volatility)
- High-Risk High-Reward: BRL, MXN, ZAR maintaining extreme volatility profiles above 0.67%

7) Currency Group Behavior

```

1 -- Analyze currencies by regions and development status
2 - WITH currency_regions AS (
3   SELECT
4     target_currency,
5     CASE
6       -- Major Developed
7       WHEN target_currency IN ('USD', 'EUR', 'GBP', 'JPY', 'CHF', 'CAD', 'AUD', 'NZD') THEN 'MAJOR_DEVELOPED'
8       -- Scandinavian
9       WHEN target_currency IN ('SEK', 'NOK', 'DKK') THEN 'SCANDINAVIAN'
10      -- Asian Developed
11      WHEN target_currency IN ('SGD', 'HKD', 'KRW') THEN 'ASIAN_DEVELOPED'
12      -- Emerging Markets
13      WHEN target_currency IN ('CNY', 'INR', 'BRL', 'RUB', 'ZAR', 'MXN', 'TRY') THEN 'EMERGING_MARKETS'
14      -- Eastern European
15      WHEN target_currency IN ('PLN', 'HUF', 'CZK', 'RON') THEN 'EASTERN_EUROPE'
16      -- Other
17      ELSE 'OTHER'
18    END AS currency_group
19  FROM currency_analytics.processed_currency_rates
20  GROUP BY target_currency
21 )
22 SELECT
23   a.currency_group AS group1,
24   b.currency_group AS group2,
25   ROUND(CORR(x.daily_return, y.daily_return), 4) AS inter_group_correlation,
26   COUNT(*) AS overlapping_pairs
27 FROM currency_regions a
28 JOIN currency_regions b ON a.currency_group <= b.currency_group
29 JOIN currency_analytics.processed_currency_rates x ON a.target_currency = x.target_currency
30 JOIN currency_analytics.processed_currency_rates y ON b.target_currency = y.target_currency
31 AND x.trade_date = y.trade_date
32 AND x.target_currency != y.target_currency
33 GROUP BY a.currency_group, b.currency_group
34 ORDER BY a.currency_group, b.currency_group;

```

#	group1	group2	inter_group_correlation	overlapping_pairs
1	ASIAN_DEVELOPED	ASIAN_DEVELOPED	0.4151	4290
2	ASIAN_DEVELOPED	EASTERN_EUROPE	-0.1106	8580
3	ASIAN_DEVELOPED	EMERGING_MARKETS	0.3629	12870
4	ASIAN_DEVELOPED	MAJOR_DEVELOPED	0.3264	15015
5	ASIAN_DEVELOPED	OTHER	0.379	15018
6	ASIAN_DEVELOPED	SCANDINAVIAN	-0.008	6435
7	EASTERN_EUROPE	EASTERN_EUROPE	0.2202	8580
8	EASTERN_EUROPE	EMERGING_MARKETS	0.0039	17160
9	EASTERN_EUROPE	MAJOR_DEVELOPED	-0.0524	20020
10	EASTERN_EUROPE	OTHER	-0.0538	20024
11	EASTERN_EUROPE	SCANDINAVIAN	0.0952	8580
12	EMERGING_MARKETS	EMERGING_MARKETS	0.2983	21450
13	EMERGING_MARKETS	MAJOR_DEVELOPED	0.2296	30030
14	EMERGING_MARKETS	OTHER	0.263	30036
15	EMERGING_MARKETS	SCANDINAVIAN	0.0586	12870
16	MAJOR_DEVELOPED	MAJOR_DEVELOPED	0.2413	30030
17	MAJOR_DEVELOPED	OTHER	0.2327	35042
18	MAJOR_DEVELOPED	SCANDINAVIAN	0.1243	15015
19	OTHER	OTHER	0.2505	30044
20	OTHER	SCANDINAVIAN	0.021	15018
21	SCANDINAVIAN	SCANDINAVIAN	0.2557	4290

- Asian Developed Block: Strong internal cohesion (0.415 correlation) with moderate ties to Emerging Markets (0.363)
- Eastern European Currencies: Show negative correlation with Asian Developed (-0.111), offering potential hedging opportunities
- Major Developed Currencies: Demonstrate balanced exposure across regions with moderate emerging market correlation (0.230)

4.2 Python EDA Engine: Advanced Statistical Analysis

This engine performs deep statistical analysis, risk modeling, and generates predictive forecasts using AWS Lambda. The analysis processed 999 records across 30 currencies from January to February 2023.

1) Key Statistical Insights

KEY STATISTICAL INSIGHTS:
- Average exchange rate: 619.787376
- Average daily return: 0.007464%
- Average 30-day volatility: 0.513986
- Most volatile currency: BRL
- Best performing currency: ZAR

2) Volatility Risk Assessment

TEXT VISUALIZATIONS GENERATED:
VOLATILITY RANKING - TOP 10 MOST VOLATILE CURRENCIES

Currency	Avg Volatility	Risk Level	Status
BRL	1.023	HIGH RISK	🚫 AVOID
JPY	0.785	HIGH RISK	🚫 AVOID
ZAR	0.778	HIGH RISK	🚫 AVOID
NOK	0.768	HIGH RISK	🚫 AVOID
AUD	0.727	HIGH RISK	🚫 AVOID
MXN	0.707	HIGH RISK	🚫 AVOID
HUF	0.686	MEDIUM RISK	⚠️ MONITOR
TRY	0.680	MEDIUM RISK	⚠️ MONITOR
USD	0.677	MEDIUM RISK	⚠️ MONITOR
HKD	0.669	MEDIUM RISK	⚠️ MONITOR

BRL (1.02% volatility) identified as highest risk, 78% more volatile than CAD

3) Performance Matrix Analysis

PERFORMANCE MATRIX - RISK vs RETURN ANALYSIS

Currency	Return %	Volatility	Sharpe	Verdict
HUF	-5.10	0.686	-0.203	🚫 AVOID
JPY	2.46	0.785	0.073	✅ GOOD
MXN	-4.12	0.707	-0.192	🚫 AVOID
NOK	3.86	0.768	0.163	🏆 TOP PERFORMER
TRY	0.82	0.680	0.051	✅ GOOD
ZAR	6.15	0.778	0.282	🏆 TOP PERFORMER

ZAR delivers best risk-adjusted returns (Sharpe ratio: 0.28)

4) Currency Correlation Clusters

CORRELATION CLUSTERS – STRONGLY LINKED CURRENCIES

- 🔗 ASIAN BLOC
Currencies: HKD, USD, INR, CNY, SGD
Strength: VERY STRONG (0.8–0.99)
- 🔗 LATAM PAIR
Currencies: BRL, MXN
Strength: MODERATE (0.50)
- 🔗 OCEANIA PAIR
Currencies: AUD, NZD
Strength: VERY STRONG (0.82)
- 🔗 SCANDINAVIAN
Currencies: NOK, SEK
Strength: MODERATE (0.53)

Asian currency bloc shows near-perfect correlation (0.8-0.99)

5) Forecasting Output

currency_tomorrow_forecasts (1)							
	currency	forecast_date	predicted_rate	current_rate	7_day_average	30_day_average	forecast_timestamp
1	USD	2025-10-20	1.070604	1.07	1.072014	1.07909	2025-10-19 22:54:59
2	INR	2025-10-20	88.636571	88.64	88.628571	88.468333	2025-10-19 22:54:59
3	AUD	2025-10-20	1.548221	1.55	1.544071	1.548577	2025-10-19 22:54:59
4	GBP	2025-10-20	0.8878	0.88883	0.885396	0.88432	2025-10-19 22:54:59
5	SGD	2025-10-20	1.426974	1.4285	1.423414	1.42799	2025-10-19 22:54:59
6	JPY	2025-10-20	142.547714	143.05	141.375714	141.022	2025-10-19 22:54:59

Next-day predictions for 6 major currencies using weighted moving averages

5.0 Tableau Visualization Layer

The platform features an interactive Tableau dashboard with six integrated worksheets, enabling deep exploration of the data.

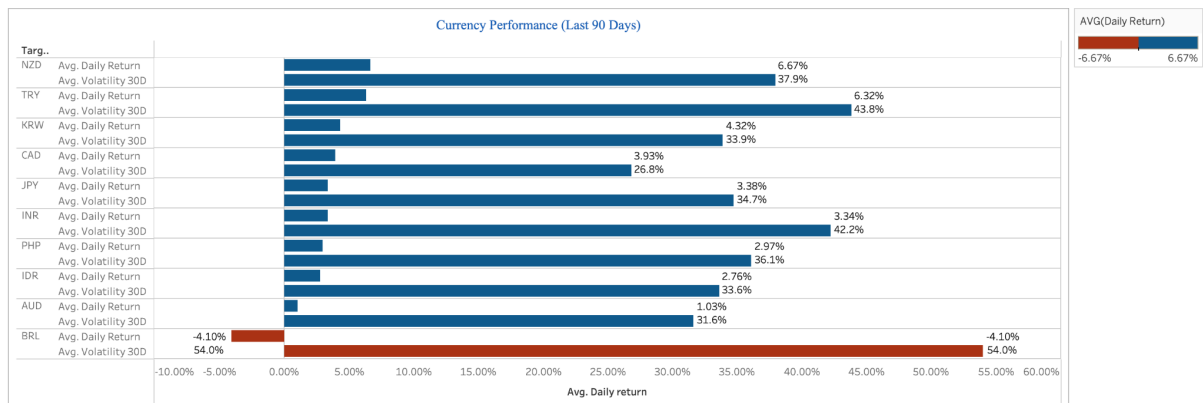
Worksheets:

1) Currency Performance Grid

Purpose: Current market snapshot

Key Metrics: Daily Return, Volatility

Insights Delivered: Top/bottom performers, at-a-glance risk assessment.

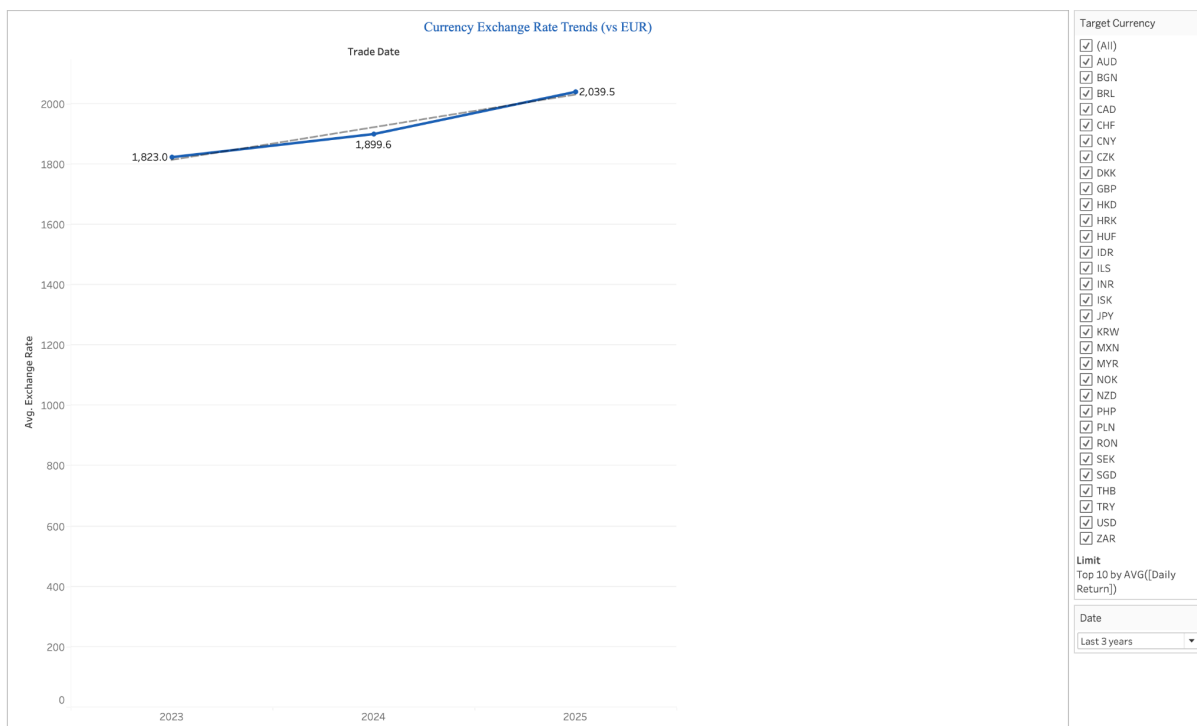


2) Exchange Rate Trends

Purpose: Historical analysis

Key Metrics: Exchange Rates over Time

Insights Delivered: Long-term patterns, support/resistance levels.

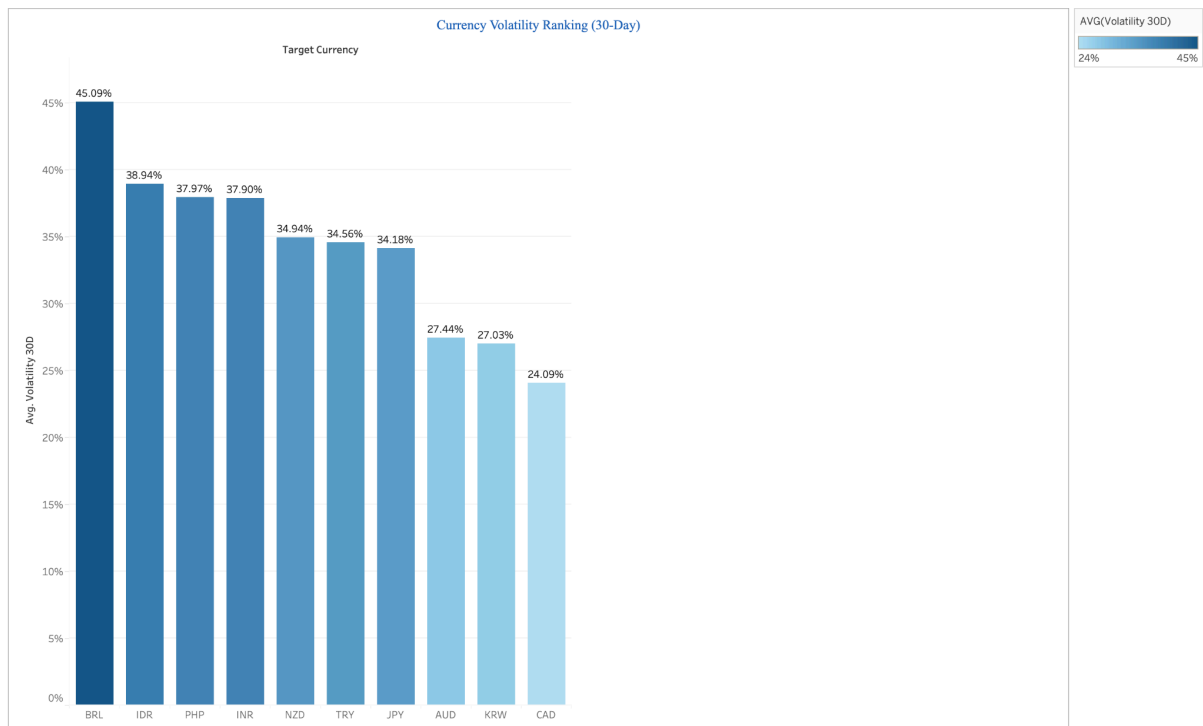


3) Volatility Ranking

Purpose: Risk assessment

Key Metrics: 30-Day Volatility

Insights Delivered: Currency risk profiles and stability rankings.

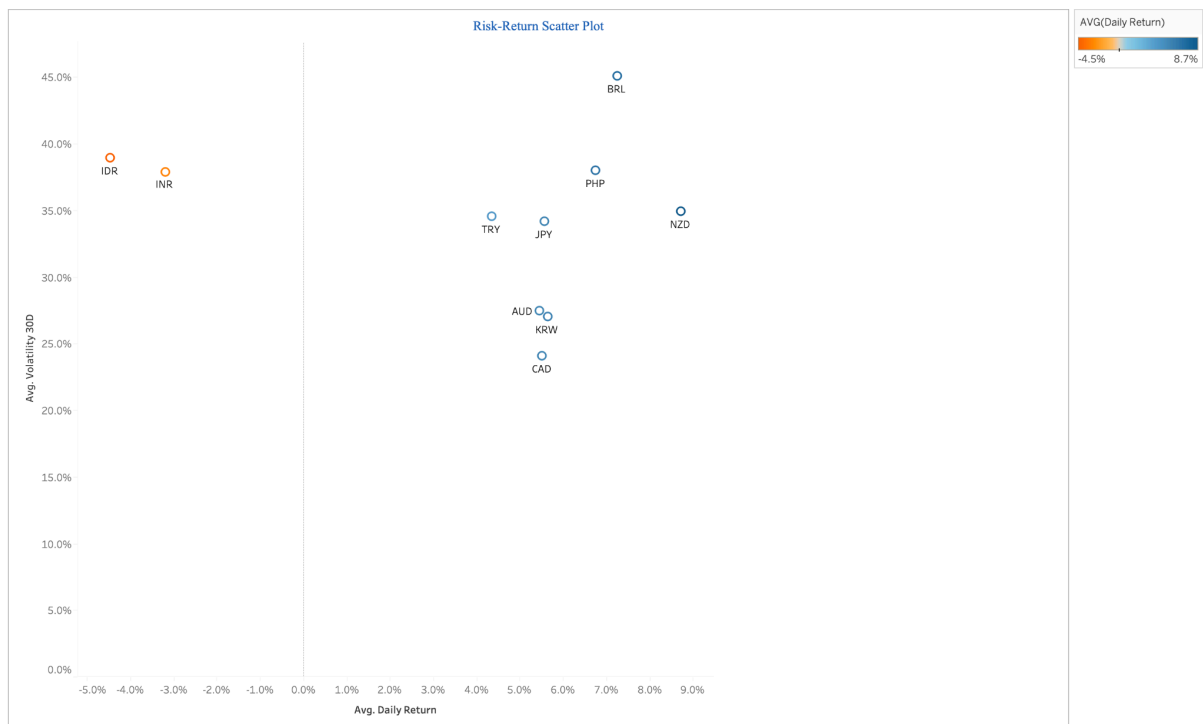


4) Risk-Return Scatter Plot

Purpose: Investment support

Key Metrics: Return vs. Volatility

Insights Delivered: Efficient frontier identification, risk-adjusted returns.

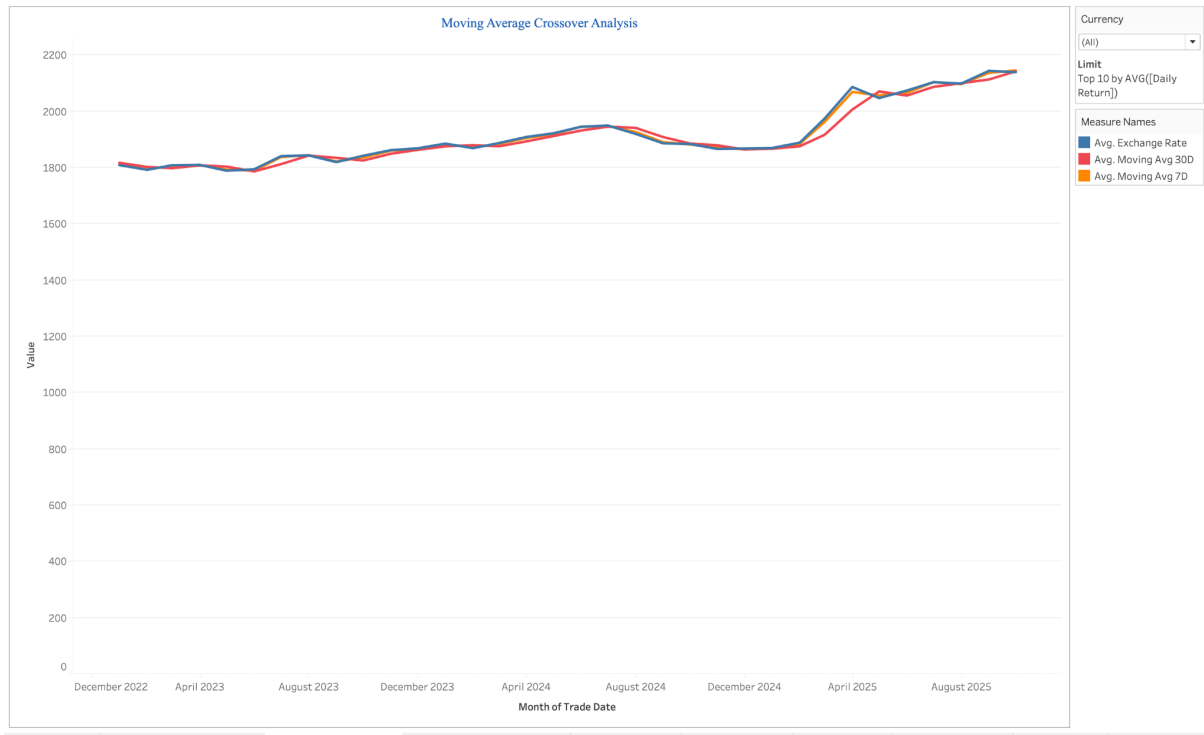


5) Moving Average Crossover

Purpose: Technical analysis

Key Metrics: 7-day vs. 30-day Averages

Insights Delivered: Trend signals, potential entry/exit points.



6) Quarterly Performance Heatmap

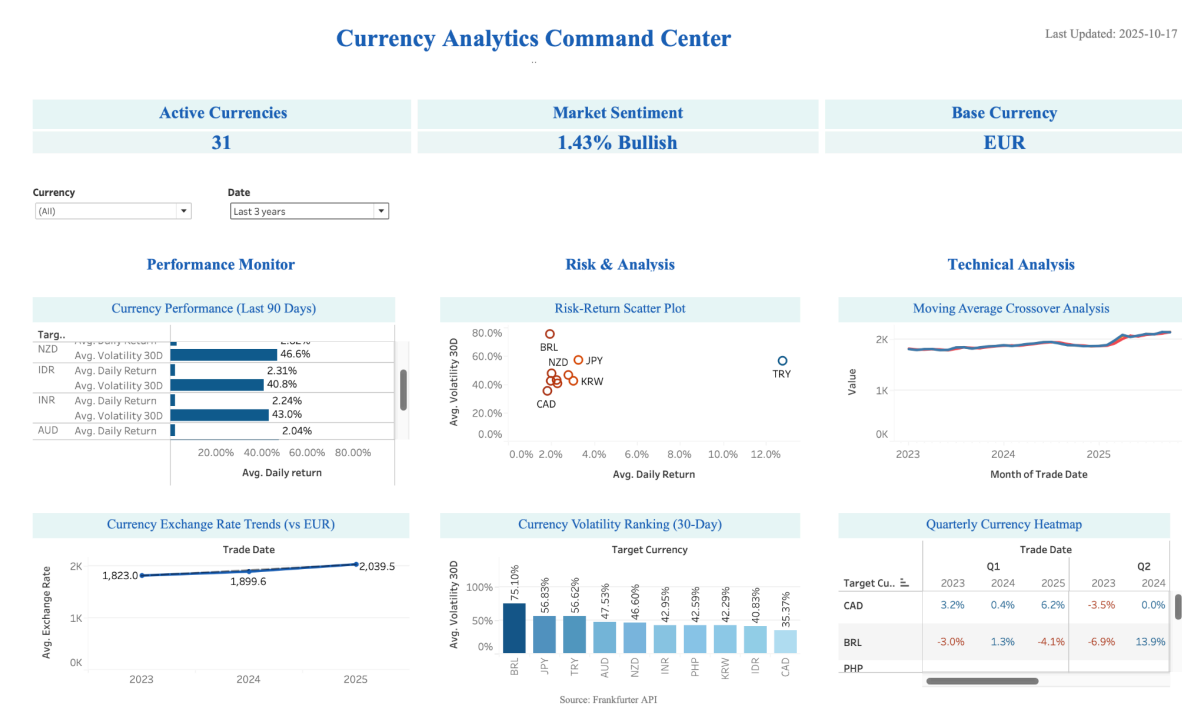
Purpose: Seasonal analysis

Key Metrics: Quarterly Returns

Insights Delivered: Seasonal trends and performance consistency.



Dashboard Features:



KPI Summary: Tracks active currencies, market sentiment, and data currentness

Interactive Filtering: Cross-filtering allows clicking on any currency or date range to update all worksheets simultaneously.

6.0 Business Value & Strategic Recommendations

6.1 Actionable Insights for Key Personas

For Risk-Averse Investors:

- 1) Focus on: CAD, SGD, DKK (low volatility, stable returns).
- 2) Avoid: TRY, BRL, MXN (extreme volatility).
- 3) Monitor: JPY (unusual behavior).

For Active Traders:

- 1) Monday Morning Strategy: Capitalize on the week's positive start.
- 2) Regional Arbitrage: Exploit correlations between Latin American and North American currencies.
- 3) Friday Protection: Implement hedging to mitigate weekend risk.

For Portfolio Managers:

- 1) Diversification Note: Geographic diversification offers limited benefits; focus on currency-specific selection.
- 2) Currency Selection: Prefer Asian developed currencies for stability.
- 3) Risk Monitoring: Implement daily volatility checks on emerging market exposures.

6.2 Final Model Outputs

- 1) Executive Dashboard (Tableau): Provides real-time metrics on market health, risk level, top performers, and data freshness.
- 2) Risk Reports (Python EDA): Delivers deep-dive assessments on volatility, unusual currency behavior, and correlation risks.
- 3) Strategic Recommendations (SQL EDA): Offers clear trading strategies based on timing, hedging pairs, and risk management limits.